Full support and positive density in the zero-temperature SK model

One lower bound, two consequences · A reconstruction of OpenAI's argument · 25 October 2026

Abstract. Every integrable minimiser of the zero-temperature Parisi functional for the pure, zero-field Sherrington–Kirkpatrick model has full Stieltjes support on $[0,1)$ and a smooth density that is strictly positive on $(0,1)$. We reconstruct two uses of the quantitative estimate in OpenAI's proof. On a positive interval of constancy, the estimate makes the derivative of a gradient moment strictly convex; variational contact at the endpoints then forces an impossible area deficit. A separate boundary estimate excludes terminal intervals. After full support gives the moment identities and smoothness, a local constant replacement transfers the same inequality to the minimiser at a fixed time and forces its density to be positive. Chen previously proved full support and smooth nonnegative density. The contribution of this account is to explain the quantitative argument, its second use and its relation to the separate quantum approximate optimization algorithm (QAOA) construction, while identifying the analytic estimates imported from the original sources.

1. The result and the mechanism

For independent standard Gaussian couplings J_{ij}, the pure, zero-field Sherrington–Kirkpatrick (SK) Hamiltonian is

$$H_n(\sigma) = n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j, \qquad \sigma \in \{-1,1\}^n.$$

A spin configuration σ assigns one sign to each of the n sites. The zero-temperature problem concerns the limiting maximum energy per spin,

$$P_* := \lim_{n\to\infty} n^{-1}\mathbb{E}\max_{\sigma} H_n(\sigma).$$

The Parisi formula expresses P_* as a minimum over nondecreasing functions. Our subject is the structure of a minimiser: its increments charge every interval of overlap scales below one, and in fact have strictly positive density at each interior scale. The proof uses one quantitative bound twice, first to exclude constant intervals and then, after an approximation, to prove pointwise positivity.

For two configurations, their overlap is $R(\sigma,\tau) = n^{-1}\sum_i \sigma_i\tau_i$. It determines the energy covariance: $\mathbb{E}[H_n(\sigma)H_n(\tau)] = nR^2/2 - 1/2$. Thus the Gaussian law of the energies is encoded by the overlaps. We use the covariance convention $\xi(t) = t^2/2$, so $\xi'' = 1$; adding one independent centred Gaussian of variance $1/2$ to every configuration's energy gives covariance $n\xi(R)$ and leaves the expected maximum unchanged. This is the normalisation of the ground-state Parisi formula of Auffinger and Chen (2017), Theorem 1. There are no higher-order interactions and no external field.

Let $\mathcal{U}$ consist of the nonnegative, nondecreasing, right-continuous functions $\gamma : [0,1) \to [0,\infty)$ satisfying $\int_0^1 \gamma(t)\,dt < \infty$. Define the locally finite Stieltjes measure by

$$\mu_\gamma([0,t]) = \gamma(t), \qquad 0 \le t < 1.$$

Thus $\mu_\gamma((s,t]) = \gamma(t) - \gamma(s)$: the measure records the increments of γ. A point mass (atom) of size $\gamma(0)$ is allowed at zero, and the total mass need not be finite or equal to one. The support consists of points whose every neighbourhood in $[0,1)$ has positive mass. A support gap is a relatively open interval of positive length with zero mass, on which γ is constant. When the measure has a density ρ, its mass on an interval is the integral of ρ there. All support statements concern this zero-temperature Stieltjes measure and are relative to $[0,1)$.

The Parisi functional $\mathcal{P}(\gamma)$ is defined using an auxiliary value function $\Phi_\gamma(t,x)$, where $x \in \mathbb{R}$ is a spatial variable and t is the overlap parameter, treated as time in the equation. Write $\Phi = \Phi_\gamma$ and use subscripts for partial derivatives. The backward equation, terminal condition and functional are

$$\Phi_t = -\tfrac{1}{2}(\Phi_{xx} + \gamma(t)\Phi_x^2), \qquad \Phi(1,x) = |x|, \qquad \mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \tfrac{1}{2}\int_0^1 t\gamma(t)\,dt.$$

The terminal value $|x| = \max\{x, -x\}$ is the zero-temperature value of one spin in a field x. For an integrable coefficient, the PDE solution is interpreted by approximation from bounded monotone coefficients. The exact control definition and stability theorem are OpenAI (2026a), §1 and Lemmas 2.1–2.2. To turn minimality of $\mathcal{P}$ into a condition on γ, we use a random process whose gradient moment measures the first-order cost of changing γ. Write $u = \Phi_x$ for the spatial gradient and $a = \Phi_{xx}$ for its spatial derivative. Let B be standard Brownian motion and define the diffusion X and the moment q by

$$dX_t = \gamma(t)u(t,X_t)\,dt + dB_t, \qquad X_0 = 0, \qquad q(t) = \mathbb{E}[u(t,X_t)^2].$$

The discrepancy $q(t) - t$ enters the first variation of $\mathcal{P}$; minimality will force $q(t) = t$ on the support of μ_γ. The diffusion exists uniquely for the fixed Brownian motion. All stochastic calculations below take place on compact time intervals ending before one, unless an endpoint limit is justified explicitly.

Auffinger, Chen and Zeng (2020), Theorem 1 proved infinite support. This need not be full support: the measure $\sum_{k\geq 1} 2^{-k}\delta_{1-2^{-k}}$ has infinitely many atoms and many gaps. Chen (2026), Theorem 1.3 subsequently proved full support and smooth nonnegative density. We follow OpenAI's quantitative proof of the support conclusion and its further strict-positivity result. The feature used twice is a lower bound valid throughout every positive constant interval of an admissible coefficient, without a minimising assumption.

Theorem 1 (full support; OpenAI (2026a), Theorem 1.1). If $\gamma \in \mathcal{U}$ minimises $\mathcal{P}$, then

$$\operatorname{supp}\mu_\gamma = [0,1), \qquad \gamma(t_2) > \gamma(t_1) \quad (0 \leq t_1 < t_2 < 1).$$

For every associated diffusion and every $0 \leq t < 1$,

$$q(t) = t, \qquad \mathbb{E}[a(t,X_t)^2] = 1.$$

Existence of a minimiser is a separate input from Auffinger and Chen (2017), Theorem 1 and the proof of Lemma 3. Integrability allows γ to grow near one. Zero field gives $u(0,0) = 0$, and $\xi'' = 1$ is used in the curvature calculation; the theorem here does not extend these hypotheses to nonzero field or a general mixed-spin covariance.

The proof compares two demands on a hypothetical gap. Minimality forces q to agree with time at its endpoints, with endpoint slopes at most one. On a positive constant interval a quantitative estimate makes q' strictly convex, so its integral is smaller than the interval's length. This contradicts the required change in q. After excluding all gaps, the identity $q' = 1$ supplies the equation from which the same estimate will force $\gamma' > 0$.

The support identities also supply the normalisation used in a proof of asymptotic optimality for the quantum approximate optimization algorithm (QAOA). Section 5 states that separate theorem and the finite Gaussian correlation quantity it uses. The density conclusion is not needed for that construction. We assume Itô calculus and martingale convergence; the nontrivial PDE and integrated estimates are identified at their uses.

2. Minimality gives contact; curvature excludes gaps

For an admissible coefficient, Φ is even, convex and one-Lipschitz in space. Its positive-order spatial derivatives are bounded and continuous on each strip $[0,T]\times\mathbb{R}$, $T<1$; the PDE holds almost everywhere in time. The compact-strip approximation theorem justifies the following Itô calculations even when γ has jumps (see OpenAI (2026a), Lemmas 2.1–2.2). The control method has antecedents in Jagannath and Tobasco (2016) at positive temperature and Chen, Handschy and Lerman (2018), Proposition 2 and Theorem 5 at zero temperature.

Fix a minimiser. Itô's formula cancels the drift of $u(t,X_t)$:

$$du(t,X_t)=a(t,X_t)\,dB_t,\qquad q(0)=0,\qquad q'(t)=\mathbb{E}[a(t,X_t)^2].$$

In particular q' is continuous across coefficient jumps. Minimality means that every allowed one-sided change has nonnegative first-order cost. The one-sided variation formula gives the rate of change of the functional when γ is replaced by $\gamma+\varepsilon\eta$ as $\varepsilon\downarrow 0$:

$$D_+\mathcal{P}(\gamma)[\eta]=\tfrac{1}{2}\int_0^1\eta(t)(q(t)-t)\,dt, \tag{1}$$

for every integrable direction η for which $\gamma+\varepsilon\eta\in\mathcal{U}$ for all sufficiently small $\varepsilon\geq 0$ (see OpenAI (2026a), Lemma 6.1). Earlier zero-temperature variation and support identities appear in Chen, Handschy and Lerman (2018), Propositions 3–4.

Adding a unit atom at s raises the cumulative function at every $t\geq s$, so its direction is $\eta=\mathbf{1}_{[s,1)}$. The first-order cost is therefore a tail integral:

$$G(s):=\int_s^1(q(t)-t)\,dt\geq 0.$$

Here $G(s)/2$ is the first-order cost of the added atom. Conversely, let $\nu=\mu_\gamma|_K$ for a compact $K\subset[0,1)$. For $0\leq\varepsilon\leq 1$, the measure $\mu_\gamma-\varepsilon\nu=(1-\varepsilon\mathbf{1}_K)\mu_\gamma$ is nonnegative, and its cumulative function is $\gamma-\varepsilon\nu([0,\cdot])\in\mathcal{U}$. Minimality gives

$$0\leq D_+\mathcal{P}(\gamma)[-\nu([0,\cdot])]=-\tfrac{1}{2}\int_K G(s)\,d\mu_\gamma(s).$$

The change of integration order is justified by $|q-t|\leq 1$ and $\mu_\gamma(K)<\infty$. Since $G\geq 0$, it vanishes almost everywhere for μ_γ, and continuity makes it vanish on $S=\operatorname{supp}\mu_\gamma$. The variation removes a fraction of all mass in K, not only atoms; no density or probability constraint is assumed.

At a support point $s>0$, G has a minimum. Hence

$$G'(s)=s-q(s)=0,\qquad G''(s)=1-q'(s)\geq 0,\quad\text{so}\quad q(s)=s,\quad q'(s)\leq 1. \tag{2}$$

If $0\in S$, contact gives $G(0)=0$; independently, zero field gives $q(0)=0$. Hence $G(h)=\frac{1}{2}(1-q'(0))h^2+o(h^2)$; nonnegativity gives the same slope bound at zero. These are the contact conditions of OpenAI (2026a), Lemma 6.2.

Lemma 2 (constant-interval curvature; OpenAI (2026a), Proposition 5.1). Let $\gamma\in\mathcal{U}$, with its associated diffusion and moment q. If $\gamma=c$ on an open interval $I\subset(0,1)$, called a plateau, then for $t\in I$

$$q'''(t)\geq\frac{c^2}{1000}\mathbb{E}[a(t,X_t)^4]>0\quad\text{if } c>0. \tag{3}$$

If $c = 0$, then $q''(t) = \mathbb{E}[\Phi_{xxx}(t, X_t)^2] > 0$.

Minimality is not a hypothesis of this lemma. Section 3 derives its moment polynomial and explains the integrated estimate imported from the source. We first use the lemma to exclude gaps. A complementary interval with two support endpoints admits two contact conditions; an initial interval without support at zero has coefficient zero; a terminal interval requires a boundary estimate.

Suppose (ℓ, r) is a support gap with endpoints in S and $r < 1$. The coefficient is constant there. If its value is positive, strict convexity of q' and (2) give

$$q'(t) < \frac{r-t}{r-\ell} q'(\ell) + \frac{t-\ell}{r-\ell} q'(r) \le 1 \qquad (\ell < t < r).$$

Integrating contradicts $q(r) - q(\ell) = r - \ell$. The geometric obstruction is an area deficit under q'. If the constant is zero, strict increase gives $q'(t) < q'(r) \le 1$ and the same contradiction. An initial gap $[0, r)$ with $0 \notin S$ and $r \in S$ has no mass, hence $\gamma = 0$ there. Its right endpoint gives $q(r) = r$ and $q'(r) \le 1$, while zero field gives $q(0) = 0$. The same integral contradiction follows without assuming $G(0) = 0$. A gap $(0, r)$ with $0 \in S$ is already covered by the two-contact case.

A terminal gap cannot use contact at one, which lies outside the support domain. Its exclusion was recorded by Auffinger, Chen and Zeng (2020), Remark 7. The quantitative argument in OpenAI (2026a), Lemma 6.3 proves, on any supposed constant interval ending at one,

$$1 - q(t) \ge C\sqrt{1-t}$$

for some $C > 0$ and all t sufficiently close to one. Two estimates produce this scale. The heat representation of the terminal value $|x|$ gives $1 - u(t,x)^2 \ge \kappa > 0$ on $|x| \le \sqrt{1-t}$. A transition-kernel comparison from a fixed earlier time gives a density $f_t(x) \ge m > 0$ on the fixed window $|x| \le 1$, uniformly as $t \uparrow 1$. Thus

$$1 - q(t) = \int_{\mathbb{R}} (1 - u(t,x)^2) f_t(x)\, dx \ge 2m\kappa\sqrt{1-t}.$$

The time-uniform density estimate, proved in Lemma 6.3, equations (6.6)–(6.7), is essential; pointwise positivity of each density would not suffice. For $h = 1 - t < C^2$, we have $q(t) - t \le h - C\sqrt{h} < 0$, so $G(s) < 0$ for s sufficiently close to one. This contradicts minimality and also excludes empty support. The zero-coefficient heat argument and late-step variation have antecedents in Chen, Handschy and Lerman (2018), Theorem 6 and its proof.

These cases exhaust the complement of the relatively closed set S. Consequently $S = [0, 1)$, contact gives $q(t) = t$, and differentiation gives $\mathbb{E}a^2 = 1$, including the right-hand value at zero. Moreover, $\gamma(t_2) - \gamma(t_1) = \mu_\gamma((t_1, t_2]) > 0$. This proves Theorem 1 using the stated analytic inputs.

3. The integrated estimate

On a plateau with $\gamma = c > 0$, put $r = c\Phi_x = cu$, $v = r_x = ca$, $w = r_{xx}$ and $z = r_{xxx}$. This rescaling makes the derivative identities polynomial. The PDE becomes $r_t = -\frac{1}{2} r_{xx} - r r_x$, and the diffusion has spatial generator $\frac{1}{2}\partial_{xx} + r\partial_x$. Differentiating in space and applying Itô's formula gives the drift and diffusion pairs $(0, v)$, $(-v^2, w)$ and $(-3vw, z)$ for r, v, w, respectively. All expectations in this section evaluate spatial functions at (t, X_t). Starting with $c^2 q' = \mathbb{E}v^2$, Itô's formula gives

$$c^2 q'' = \mathbb{E}[w^2 - 2v^3], \qquad c^2 q''' = \mathbb{E}[z^2 - 12vw^2 + 6v^4]. \tag{4}$$

For example, the drifts of w^2 and v^3 are $z^2 - 6vw^2$ and $3vw^2 - 3v^4$, respectively. The negative mixed term is the obstacle. Convexity, which gives $v \geq 0$, does not control it.

The required lower bound holds after integration against the law of X_t. We derive the moment polynomial (4) here; the shape comparisons and rational identity yielding its lower bound are imported from OpenAI (2026a), §§3–4. They are first established for finite-step coefficients: on each constant step the equation can be transformed into a heat equation, making the needed shape restrictions accessible. We then pass to general integrable coefficients by approximation. Begin with a positive nondecreasing finite-step coefficient γ_{step}: a function constant on each interval of a finite partition. Replace the terminal value $|x|$ by the smooth approximation $M^{-1}\log(2\cosh Mx)$, with $M \geq \|\gamma_{\text{step}}\|_\infty$. On a step of height c, the exponential $e^{c\Phi}$ solves the backward heat equation. At positive time, write the diffusion density as $f = e^{c\Phi}g$. This substitution turns its Fokker–Planck equation into the forward heat equation for g. For $s = -\partial_x \log g$, the source proves joint shape restrictions: on $x > 0$, r and s are increasing and concave, $w/(2r) + v \leq 0$ is nondecreasing, and

$$j := s_x - (s/r)v = r\,\partial_x(s/r) \geq 0.$$

The last inequality couples the backward gradient to the forward density; separate monotonicity and concavity do not supply it. The ratios extend smoothly through zero because r, s, w are odd and $r_x(0) > 0$. These are analytic conclusions of OpenAI (2026a), §3, not consequences of the polynomial in (4).

The factorisation of the density gives $f_x/f = r - s$. For a suitable expression F in the shape quantities, define

$$\mathcal{W}(F) = r\partial_x F + (v + r^2 - rs)F.$$

Then $f\mathcal{W}(F) = \partial_x(rfF)$. The smooth extensions avoid an interior boundary term at zero, and the source's differentiated Gaussian tail estimates give $rfF \to 0$ at both spatial infinities for the polynomial correction used below. Hence $\mathbb{E}_f\mathcal{W}(F) = 0$, where $\mathbb{E}_f$ denotes integration against $f(x)\,dx$.

The remaining step is algebra under these analytic restrictions. OpenAI (2026a), Lemma 4.1 gives an explicit polynomial F_2 for which subtracting $\mathcal{W}(F_2)$ from $z^2 - 12vw^2 + 6v^4$ leaves a sum of weighted squares, nonnegative products and a polynomial with positive coefficients. For example, one square has the nonnegative weight $r^2 j$; this is where the joint comparison enters. The remainder's v^4 coefficient is $1661507/1521000000 > 1/1000$. Integration therefore gives

$$\int_{\mathbb{R}} (z^2 - 12vw^2 + 6v^4) f\,dx \geq \frac{1}{1000}\int_{\mathbb{R}} v^4 f\,dx. \tag{5}$$

We use (5) as an imported analytic estimate. The full polynomial and its decomposition are in Lemma 4.1 and Table 1; Proposition 4.2 integrates them using the signs and zero boundary flux supplied by Lemma 3.4. The long record, §4.4, "How the signs make a correction with zero expectation useful", reproduces those definitions and the complete identity. A pointwise sign assertion for arbitrary v, w, z would be false.

To obtain Lemma 2 for an integrable coefficient, choose positive monotone step coefficients γ_n retaining the plateau near a fixed interior time, with $\|\gamma_n - \gamma\|_1 \to 0$, and terminal parameters $M_n \geq \|\gamma_n\|_\infty$ with $M_n \to \infty$. Spatial derivatives through order four converge uniformly on a fixed strip below one, and the diffusions driven by the same Brownian motion converge (see OpenAI (2026a), Lemmas 2.1–2.2 and Proposition 5.1). Only v, w, z remain in (5), so no convergence

theorem for the auxiliary logarithmic derivative s is needed. Dividing by c^2 gives (3); strictness follows from $\Phi_{xx} > 0$ on the plateau (see OpenAI (2026a), Lemma 2.3). The zero plateau has the separate heat identity in Lemma 2. Its strictness uses a positive diffusion density and the fact that Φ_{xxx} cannot vanish identically: an even quadratic profile would contradict the one-Lipschitz bound and $|x| + O(1)$ growth.

The exponential change of variables that converts each constant-step equation to a heat equation (the Cole–Hopf representation), together with the rules joining successive steps, has antecedents in Lopatto (2026), Propositions 3.6, 7.1 and 9.1 and Chen (2026), §§3–4. The common heat and density methods should be distinguished from the strength of the resulting estimate. Under the minimiser convention of Chen's §3.1, Proposition 4.2 proves the crossing implication $q''(t) = 0 \Rightarrow q'''(t) > 0$ on a hypothetical positive gap of the minimiser. In contrast, (3) is quantitative and holds throughout every positive plateau of any admissible coefficient. Chen's Appendix C distinguishes earlier shared estimates from the parallel crossing proofs.

4. A second use: the density cannot vanish inside the interval

Full support does not force pointwise positive density. For instance, $\gamma(t) = \int_0^t (s - \frac{1}{2})^2 \, ds$ has full Stieltjes support but $\gamma'(\frac{1}{2}) = 0$: every interval of positive length has a positive integral, although the integrand vanishes at one point. The quantitative remainder in (5) rules this out for a minimiser. First, the identity $\mathbb{E}a^2 = 1$ expresses γ as a quotient of continuous moments and then gives smoothness. Only afterwards do we transfer (5) from a locally constant comparison coefficient to obtain $\gamma' > 0$.

Corollary 3 (smooth positive density; OpenAI (2026a), Corollary 6.4). Under the hypotheses of Theorem 1, $\gamma(0) = 0$ and $\gamma \in C^\infty([0,1))$, with derivatives at zero taken from the right. Its Stieltjes measure is $\mu_\gamma(dt) = \gamma'(t)\, dt$, and

$$2\gamma'(t)\mathbb{E}[a(t, X_t)^3] \geq \frac{\gamma(t)^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \qquad (0 < t < 1). \tag{6}$$

Only nonnegativity of the density at zero is asserted.

Write $b = \Phi_{xxx}$ and $d = \Phi_{xxxx}$; all spatial factors in expectations below are evaluated at (t, X_t). Since $da = -\gamma a^2\, dt + b\, dB$, Itô's formula applied to a^2 gives, for almost every t, $0 = \frac{d}{dt}\mathbb{E}a^2 = \mathbb{E}b^2 - 2\gamma\mathbb{E}a^3$. Here the left side vanishes because full support gives $\mathbb{E}a^2 = 1$. Solving for the coefficient yields

$$\gamma(t) = \frac{\mathbb{E}b^2}{2\mathbb{E}a^3}. \tag{7}$$

Initially (7) holds almost everywhere. Its denominator is at least two, because $a \geq 0$ and $\mathbb{E}a^3 \geq (\mathbb{E}a^2)^{3/2} = 1$. The moments are continuous; right continuity of γ extends (7) to every time and proves continuity of γ. Evenness gives $b(0,0) = 0$, hence $\gamma(0) = 0$.

For smoothness, fix $T < 1$ and consider all polynomial moments

$$M_P(t) = \mathbb{E}P(\Phi_x(t, X_t), \ldots, \partial_x^m \Phi(t, X_t)),$$

where P is a polynomial in any finite number m of variables. Spatial differentiation of the PDE and Itô's product rule give polynomials Q_P, R_P, involving finitely many further spatial derivatives, such that

$$M_P(t) - M_P(0) = \int_0^t [M_{Q_P}(s) + \gamma(s) M_{R_P}(s)]\, ds.$$

No derivative of γ enters this identity. Bounded spatial derivatives on the fixed strip make every moment continuous. The integral identity then makes all moments C^1, and (7) makes γ C^1. Repetition proves smoothness. This is the quotient-and-moment argument of Chen (2026), Proposition 4.6 and Appendix B.6, using the polynomial-moment method of Auffinger and Chen (2015), Lemma 2 and the proof of Theorem 2(ii). The explicit polynomials and the passage from initially nonsmooth coefficients are in OpenAI (2026a), proof of Corollary 6.4 and the long record, §5.1.

Now the two differentiated moments are

$$\frac{d}{dt}\mathbb{E}b^2 = \mathbb{E}[d^2 - 6\gamma ab^2], \qquad \frac{d}{dt}\mathbb{E}a^3 = \mathbb{E}[3ab^2 - 3\gamma a^4].$$

Differentiating $\mathbb{E}b^2 = 2\gamma\mathbb{E}a^3$ gives

$$2\gamma'\mathbb{E}a^3 = \mathbb{E}[d^2 - 12\gamma ab^2 + 6\gamma^2 a^4]. \tag{8}$$

The spatial polynomial from (4) has returned. We need the integrated inequality underlying that calculation: applying the conclusion $q^{(3)} > 0$ directly to the minimiser would be invalid, since $q(t) = t$. We transfer the spatial moments at a fixed time instead.

Fix $t \in (0,1)$ and $c = \gamma(t) > 0$. For $0 < \delta < \min(t, 1-t)$ define

$$\gamma_\delta(s) = \begin{cases} c, & t-\delta \le s < t+\delta, \\ \gamma(s), & \text{otherwise.} \end{cases}$$

The inequalities $\gamma(t-\delta) \le c \le \gamma(t+\delta)$ make both joins nondecreasing. Thus $\gamma_\delta \in \mathcal{U}$; it need not minimise $\mathcal{P}$. Moreover,

$$\|\gamma_\delta - \gamma\|_1 \le 2\delta\,[\gamma(t+\delta) - \gamma(t-\delta)] \longrightarrow 0.$$

Use a subscript δ for its value-function derivatives and diffusion. Lemma 2 and (4), already extended to integrable coefficients, give at the fixed time t

$$\mathbb{E}[d_\delta^2 - 12c\,a_\delta b_\delta^2 + 6c^2 a_\delta^4] \ge \frac{c^2}{1000}\mathbb{E}[a_\delta^4]. \tag{9}$$

Choose a single T with $t < T < 1$. Lemma 2.1 of OpenAI (2026a) gives uniform convergence in space and uniform bounds for $a_\delta, b_\delta, d_\delta$ on $[0,T]$, with the next spatial derivatives bounded as well. Lemma 2.2 couples the diffusions using the same Brownian motion and gives $\sup_{s\le T}|X_s^\delta - X_s| \to 0$. The derivative bounds control evaluation on these paths, so the integrands in (9) converge and are uniformly bounded. Dominated convergence passes their expectations to the limit. The common coefficient L^1 bound and the fixed distance from time one determine these estimates; the shrinking plateau width does not. Combining the limiting inequality with (8) proves (6). Strictness follows from $c > 0$ and $\mathbb{E}a^4 \ge (\mathbb{E}a^2)^2 = 1$. In particular, Cauchy–Schwarz gives

$$\mathbb{E}a^3 \le \sqrt{\mathbb{E}a^2\,\mathbb{E}a^4} = \sqrt{\mathbb{E}a^4} \le \mathbb{E}a^4,$$

so the same inequality implies $\gamma'(t) \ge \gamma(t)^2/2000$ for $0 < t < 1$.

We have passed spatial moments, not the time derivatives q_δ'''. Indeed the limiting minimiser has $q(t) = t$ and $q''' = 0$; the coefficient derivative in (8) is exactly what the constant-interval formula omits. The long record, §§5.1–5.2, retains the smoothness argument and fixed-time transfer.

5. The scalar consequence and a separate circuit theorem

Set $M_t = u(t, X_t)$, the bounded gradient martingale satisfying $dM_t = a(t, X_t)\, dB_t$. Full support gives $\mathbb{E}M_t^2 = t$ and $\mathbb{E}a(t, X_t)^2 = 1$: each increment $M_t - M_s$ has variance $t - s$ for $0 \le s < t < 1$. This is the normalisation used in the discrete construction below. The first moment of a instead measures the covariance gained with the driving Brownian motion, and recovers the value:

$$P_* = \mathbb{E}[UB_1] = \int_0^1 \mathbb{E}[a(t, X_t)]\, dt. \tag{10}$$

Here U is the almost-sure and L^2 limit of M_t. Boundedness gives $|U| \le 1$, while $\mathbb{E}M_t^2 \to 1$ gives $\mathbb{E}U^2 = 1$; therefore $U^2 = 1$ almost surely. To see the cancellation behind (10), the terminal PDE estimate gives $UX_1 = |X_1|$, while Itô's formula gives $\mathcal{P}(\gamma) = \mathbb{E}|X_1| - \int_0^1 t\gamma(t)\, dt$. Multiplying $X_1 = B_1 + \int_0^1 \gamma(t) M_t\, dt$ by U cancels that integral because $\mathbb{E}[UM_t] = t$. The Parisi formula then gives (10). The endpoint limits and stochastic-integral covariance are proved in OpenAI (2026a), Corollary 7.1 and the long record, §6, "Recovering the value from a martingale". This argument uses full support, not the density corollary.

The gradient-martingale and value methods have antecedents in Montanari, preprint v2, Lemmas 3.2–3.4, under its positive-temperature support assumption, and in the zero-temperature calculations of El Alaoui, Montanari and Sellke, preprint v1, Lemmas 6.5 and 6.13.

The circuit proof needs a finite correlation construction as well as the scalar equality. For fixed $0 < T < 1$, OpenAI (2026a), Proposition 7.2 constructs, for all sufficiently large N, from independent standard normals $Z_1, \dots, Z_{N+1}$, a sign-valued function F_N and normalised increments $g_j = h_j(Z_1, \dots, Z_{j-1})Z_j$, $1 \le j \le N$, with $\mathbb{E}g_j^2 = 1$, such that

$$\lim_{N\to\infty} \sum_{j=1}^{N} \mathbb{E}[F_N g_j]\, \mathbb{E}[F_N Z_{j+1}] = \int_0^T \mathbb{E}[a(t, X_t)]\, dt. \tag{11}$$

The multipliers h_j are normalised Hessians along the Euler approximation to X. The extra variable Z_{N+1} rounds its terminal gradient to a sign while preserving the conditional mean, and hence its correlations with the preceding Gaussian variables. With $\Delta = T/N$, the correlations with g_j have leading term $\sqrt{\Delta}$, and the first N correlations with Z_i have leading term $\sqrt{\Delta}$ times the expected Hessian at the preceding Euler time. Both vectors have bounded ℓ^2 norm and their errors tend to zero in ℓ^2, so the errors vanish after pairing. The products with $j < N$ in (11) then form a Riemann sum for $\int_0^T \mathbb{E}a$; the remaining $j = N$ term, involving the rounding coordinate, vanishes. The mesh limit is taken at fixed T; only then does $T \uparrow 1$ give P_*. The construction and its error estimates are in the long record, §7, "A finite Gaussian spin and its adjacent correlations".

QAOA alternates evolution under the cost Hamiltonian with transverse-field mixing, starting from the uniform superposition of spin configurations Farhi, Goldstone and Gutmann (2014). Each pair of operations is a layer, and their durations are its angles. For every $\varepsilon > 0$, OpenAI (2026b), Theorem 1.1 constructs a finite depth and deterministic angles, independent of n and the couplings J_{ij}, whose limiting disorder-averaged expected energy per spin is at least $P_* - \varepsilon$. The depth and angles may depend on ε; $n \to \infty$ is taken with them fixed. Farhi, Goldstone, Gutmann and Zhou (2022) established the fixed-angle thermodynamic limits in this convention. The eventual-optimality conclusion was conjectured by Basso, Farhi, Marwaha, Villalonga and Zhou (2022), §7, equation (7.2).

The additional circuit argument must realise the correlation value in (11). Its recursive construction, gate approximation, removal of auxiliary randomness and transfer to finite SK systems are separate steps, reconstructed in the long record, §9, "From the scalar formula to one ordinary QAOA word", from OpenAI (2026b), §§2–6 and Appendix A. The selected argument gives no quantitative depth bound, finite-size rate or efficient procedure for finding the angles.

6. Source and formal scope

This AI-assisted article reconstructs existing results. The ordinary analytic arguments are in the cited manuscripts; the integrated estimate (5), its shape hypotheses and the compact-strip approximation are substantial imported inputs.

The selected full-support formal statement asserts the support and moment conclusions conditional on an admissible minimiser, and constructs its associated diffusion. A separate scalar formal statement covers value identities and finite Gaussian approximation, with minimiser and diffusion premises for those clauses. The two formal models have not been transported into one another in this record. Neither statement includes the positive-density corollary or QAOA convergence. The corresponding full-support and scalar implementations use Lean and mathlib.

No completed research-target acceptance by Comparator is recorded for this edition. File-integrity checks and exported theorem specifications do not establish acceptance of their implementations. Exact versions, build attempts and the correspondence with the mathematical statements are retained in the companion source review and the long record, §8, "What the formal statements establish", and §11, "Source checks and finite calculation".

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